

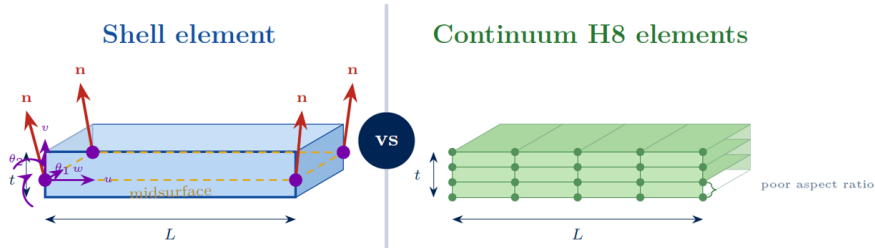
Finite Element Method

Structural Elements
MECH4103 — Spring 2026

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	Shell element	Continuum H8
DOF / node	5 ($u, v, w, \theta_1, \theta_2$)	3 (u, v, w)
Thickness treatment	analytical (Kirchhoff/Mindlin)	≥ 3 elements required
Aspect ratio issue	none	severe when $t \ll L$
Best suited for	$t/L \ll 1$ (thin shells)	$t/L \sim 1$ (bulky parts)

- A very useful group of elements in finite element analysis is beam and shell elements.
- These elements are used to model structures and components that are **thin relative to their major dimensions**.
- Some examples where beam and shell elements are applied:
 - Beams and columns of high-rise structures.
 - The sheet metal and frame of various vehicles, which are usually modelled by shell elements.
- They are thin compared to their other dimensions so that the distribution of strain through the thickness (and hence the form of the displacement field) takes a very simple form.
- For these applications, many solid elements would be needed to obtain a reasonable solution, as solid elements must have a reasonable aspect ratio to be accurate.
- So for a thin member a **huge number of solid elements** would be needed.

- ▶ Previous lectures: **continuum elements** (T3, Q4, H8) — full 2D/3D solids.
- ▶ They require many elements through the thickness to capture bending $\Rightarrow$ computationally expensive for thin structures.

Model type	DOFs	Captures bending?	Cost for thin structures
Continuum (H8, T4)	u_x, u_y, u_z	Yes (fine mesh needed)	Very high
Beam/Plate/Shell	w, θ (beam) or w, θ_x, θ_y (plate/shell)	Yes	Low
Plane stress/strain	u_x, u_y only	No	Low (limited to in-plane loading)

Key takeaway

- ▶ Beam/plate/shell are *dimensionally reduced* — they exploit thinness with kinematic assumptions (normals remain straight).
- ▶ Plane stress/strain are also 2D but **cannot** model transverse bending — only membrane (in-plane) deformations.
- ▶ For bending of thin structures: beam/plate/shell are the **right (efficient)** choice.

- In this chapter we perform the static analysis of **Bernoulli beams** under bending loads.
- Consider a beam in the xy plane, with constant cross-section area A .

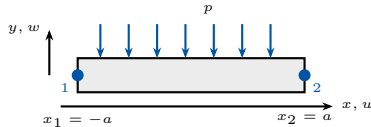
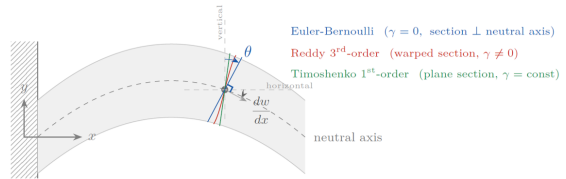


Figure. Bernoulli beam element with two nodes.

$$\gamma = \theta - \frac{dw}{dx}$$

θ = cross-section rotation from **vertical**; dw/dx = slope of neutral axis from **horizontal**



Notation

u, w = axial & transverse displacement θ = cross-section rotation $\gamma = \theta - dw/dx$ = shear angle y = coord. from neutral axis h = beam half-thickness

Euler-Bernoulli

$$u = -y \frac{dw}{dx}$$

$\theta \equiv dw/dx$, hence $\gamma = 0$ (section $\perp$ neutral axis, no shear deformation)

Timoshenko (1st-order)

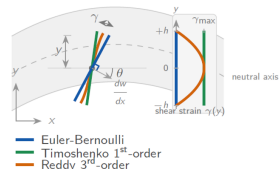
$$u = -y \theta$$

θ independent of dw/dx ; $\gamma = \theta - dw/dx = \text{const}$ (requires shear correction k)

Reddy (3rd-order)

$$u = -y \left[\theta - \frac{4y^2}{3h^2} \left(\theta + \frac{dw}{dx} \right) \right]$$

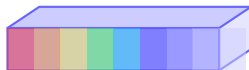
Parabolic shear: $\gamma(y) = (1 - 4y^2/3h^2) \gamma_0$; $\gamma = 0$ at $y = \pm h$ (no correction k needed)



- For the Euler-Bernoulli beam model to be applicable, the beam must be **slender**, meaning its length is significantly greater than its cross-sectional dimensions.
- A commonly accepted **length-to-diameter aspect ratio** (L/D , where L is the length and D is the diameter or characteristic cross-sectional dimension) is:

$$\frac{L}{D} \geq 10$$

- This ensures the beam is long enough relative to its thickness to neglect shear deformation and rotary inertia, which are key assumptions in the Euler-Bernoulli model.
- For ratios below 10, shear effects become significant, and the **Timoshenko beam model** may be more appropriate.



3D solid mesh



Beam element model

- In this chapter we perform the static analysis of Bernoulli beams under bending loads.
- Consider a beam in the xy plane, with constant cross-section area A .
- The Euler-Bernoulli beam theory assumes that **undeformed plane sections remain plane** under deformation.

$$\theta = \frac{\partial w}{\partial x}, \quad u = -y \frac{\partial w}{\partial x}$$

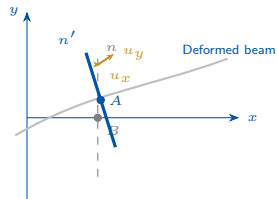


Figure. Bernoulli beam element with two nodes.

For conservative structural problems (e.g., Euler–Bernoulli beam):

- ① **Minimum Potential Energy (MPE)** — global energy minimization
- ② **Principle of Virtual Work (PVW)** — integral equilibrium statement
- ③ **Weak Form (Galerkin)** — weighted-residual / mathematical approach

All three lead to the same discrete equations:

$$[K^e]\{w^e\} = \{F^e\}$$

Why show them together?

- ▶ MPE requires a potential (energy functional) to exist. Limited to conservative, self-adjoint problems (e.g., linear elasticity, beams, heat conduction with constant properties). Not applicable to problems like fluid flow with convection and non-conservative loads.
- ▶ PVW is more general than MPE. It works for non-conservative loads and dynamics. Still relies on the concept of virtual displacements and is inherently mechanical/physical.
- ▶ Weak form is the most general. It directly starts from the strong form (PDE) and multiplies by test functions, integrating by parts. It does not require an energy functional, virtual work interpretation, or even a physical analogy.

Strong form (expanded, constant EI):

$$EI \frac{d^4 w}{dx^4} = p(x) \quad \text{on } [-a, a], \text{ with BCs.}$$

1. Minimum Potential Energy (MPE)

$$\Pi(w) = \frac{1}{2} \int_{-a}^a EI (w'')^2 dx - \int_{-a}^a p w dx$$

Set first variation $\delta \Pi = 0 \Rightarrow$

$$\int EI w'' \delta w'' dx = \int p \delta w dx$$

2. Principle of Virtual Work (PVW)

Multiply strong form by virtual displacement δw , integrate over domain, integrate by parts twice:

$$\int_{-a}^a \left(EI \frac{d^4 w}{dx^4} \right) \delta w dx = \int_{-a}^a p \delta w dx \quad \Rightarrow \quad \int_{-a}^a EI w'' (\delta w)'' dx = \int_{-a}^a p \delta w dx$$

(Boundary terms omitted for simplicity.)

3. Weak Form (Galerkin)

Multiply strong form by test function v , integrate by parts:

$$\int_{-a}^a EI w'' v'' dx = \int_{-a}^a p v dx \quad \forall v$$

All yield the same variational equation.

- The elastic strain energy is obtained as

$$U = \frac{1}{2} \int_{-a}^a EI_z \left(\frac{\partial^2 w}{\partial x^2} \right)^2 dx \quad (1)$$

where I_z is the second moment of area. The external virtual work for this element is

$$\delta W = \int_{-a}^a p \delta w dx \quad (2)$$

- At each node we consider two degrees of freedom w and dw/dx :

$$\mathbf{w}^{eT} = \begin{bmatrix} w_1 & \frac{dw_1}{dx} & w_2 & \frac{dw_2}{dx} \end{bmatrix} \quad (3)$$

- The transverse displacement is interpolated by **Hermite shape functions**:

$$w = \mathbf{N}(\xi) \mathbf{w}^e \quad (4)$$

Or in matrix form:

$$w = \begin{bmatrix} N_{u1} & N_{\theta1} & N_{u2} & N_{\theta2} \end{bmatrix} \begin{bmatrix} w_1 \\ \frac{dw_1}{dx} \\ w_2 \\ \frac{dw_2}{dx} \end{bmatrix} \quad (5)$$

- The Hermite polynomials provide C^1 continuity.

- The shape functions are defined as $(-1 \leq \xi \leq 1)$:

$$\begin{aligned} N_{u1} &= \frac{1}{4}(1 - \xi)^2(2 + \xi) \\ N_{\theta 1} &= \frac{a}{4}(1 - \xi)^2(1 + \xi) \\ N_{u2} &= \frac{1}{4}(1 + \xi)^2(2 - \xi) \\ N_{\theta 2} &= \frac{a}{4}(1 + \xi)^2(\xi - 1) \end{aligned} \quad (6)$$

- These shape functions have the following **Kronecker delta** properties:

$$N_{uI}(x_J) = \delta_{IJ}, \quad \frac{dN_{\theta I}}{dx}(x_I) = \delta_{IJ} \quad (7)$$

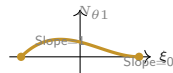
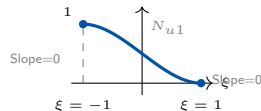
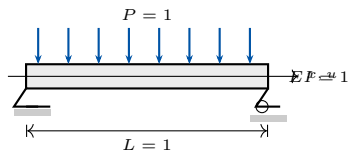


Figure. Hermite C^1 shape functions for a two-node beam.

Simply-supported Bernoulli problem

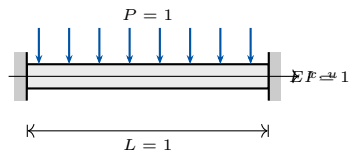


Boundary conditions

$$w(0) = w(L) = 0$$

$$\delta_{\text{exact}} = \frac{5PL^4}{384EI}$$

Clamped Bernoulli problem



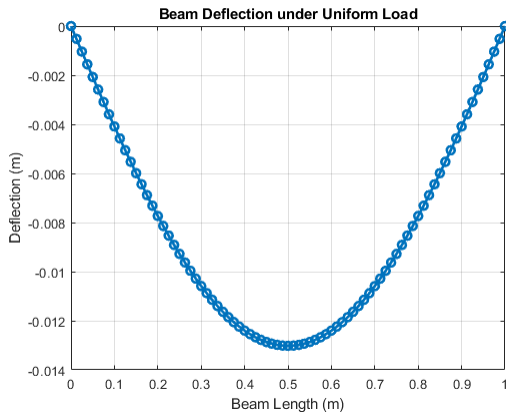
Boundary conditions

$$w(0) = w(L) = 0$$

$$\left. \frac{dw}{dx} \right|_{x=0} = \left. \frac{dw}{dx} \right|_{x=L} = 0$$

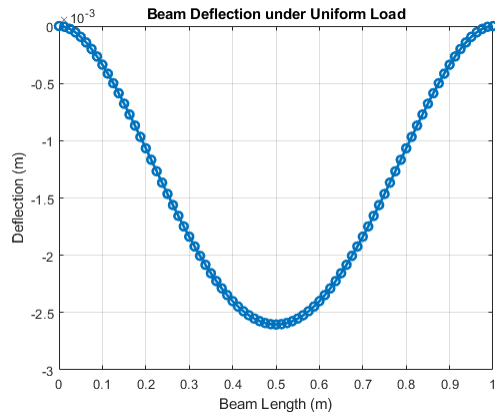
$$\delta_{\text{exact}} = \frac{PL^4}{384EI}$$

Simply-supported Bernoulli problem



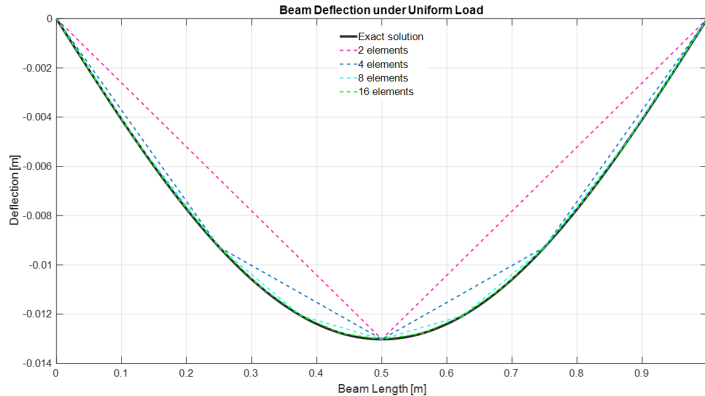
- The maximum transverse displacement is **0.0130208**, which matches the analytical solution.

Clamped-clamped Bernoulli problem



- The maximum transverse displacement for the clamped beam also matches the analytical solution.

- We want to describe some general methods for quantifying the error in a finite element solution.
- For these purposes, an exact solution is needed, but as we will see later, such exact solutions can usually be constructed by 'manufacturing' the solution.



Beam deflection under uniform load: exact solution vs. 2, 4, 8, and 16 elements.

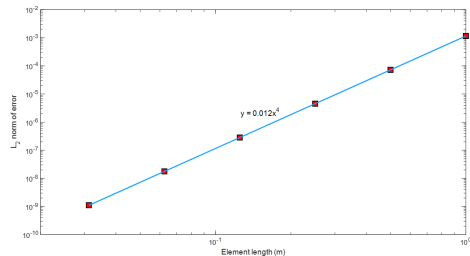
- The figure shows the $\log$ of the error norm as a function of $\log h$ (element size).
- The $\log$ of the error varies **linearly** with element size.
- If we denote the slope by α , then the L_2 error norm is:

$$\log(\|e\|_{L_2}) = C + \alpha \log h \quad (13)$$

Taking the power of both sides:

$$\|e\|_{L_2} = Ch^\alpha \quad (14)$$

where C is an arbitrary constant and α is the rate of convergence.



L_2 norm of error vs. element length (log-log). Slope = 4 confirms fourth-order convergence.

- For a cubic Hermite beam element (see Eq. (6)), $\alpha = 4$.
- **The error for the Hermite beam element is of fourth order.**
- The crucial concept: how the error decreases with element size.
- If the element size is halved, the error decreases by a **factor of 16** for cubic elements.
- **Generalization:** if a finite element contains the complete polynomial of order p , then the error in the L_2 norm of the displacement varies according to

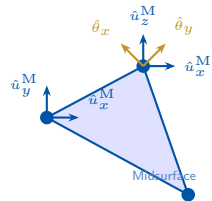
$$\|e\|_{L_2} = Ch^{p+1} \quad (15)$$

Hermite shape functions (reminder)

$$N_{u1} = \frac{1}{4}(1 - \xi)^2(2 + \xi), \quad N_{\theta1} = \frac{a}{4}(1 - \xi)^2(1 + \xi), \quad N_{u2} = \frac{1}{4}(1 + \xi)^2(2 - \xi), \quad N_{\theta2} = \frac{a}{4}(1 + \xi)^2(\xi - 1)$$

- Shells can be viewed as **generalizations of beams** to curved surfaces in three-dimensional space.
- The easiest way to describe shell elements is to consider the triangular flat facet shell element.
- We consider the counterpart of Euler–Bernoulli theory, which is called **Kirchhoff–Love shell theory**.
- The central assumption of this shell theory is identical to that in Euler–Bernoulli beam theory: *normals to the midsurface remain straight and normal*.
- In the coordinate system tangent to the midsurface, the displacements can be written in terms of the rotations:

$$\hat{u}_x = \hat{u}_x^M - \hat{z} \hat{\theta}_y, \quad \hat{u}_y = \hat{u}_y^M + \hat{z} \hat{\theta}_x, \quad \hat{u}_z = \hat{u}_z^M(\hat{x}, \hat{y})$$



where $\hat{u}_x^M$ and $\hat{u}_y^M$ are the x and y components of the midsurface displacements. From the assumption that normals remain normal, it follows that

$$\hat{\theta}_x = -\frac{\partial \hat{u}_z}{\partial \hat{y}}, \quad \hat{\theta}_y = \frac{\partial \hat{u}_z}{\partial \hat{x}}. \quad (16)$$

From the strain–displacement equations in three dimensions, we obtain

$$\begin{aligned} \hat{\epsilon}_{xx} &= \frac{\partial \hat{u}_x^M}{\partial \hat{x}} - \hat{z} \frac{\partial^2 \hat{u}_z}{\partial \hat{x}^2}, & \hat{\epsilon}_{yy} &= \frac{\partial \hat{u}_y^M}{\partial \hat{y}} + \hat{z} \frac{\partial^2 \hat{u}_z}{\partial \hat{y}^2}, \\ \hat{\gamma}_{xy} &= \frac{\partial \hat{u}_x^M}{\partial \hat{y}} + \frac{\partial \hat{u}_y^M}{\partial \hat{x}} - 2\hat{z} \frac{\partial^2 \hat{u}_z}{\partial \hat{x} \partial \hat{y}}, \\ \hat{\epsilon}_{zz} &= \hat{\gamma}_{xz} = \hat{\gamma}_{yz} = 0. \end{aligned} \quad (17)$$

- It can be seen that the strains in a shell have the same basic character as in a beam.
- They vary **linearly through the thickness** of the shell and consist of strains due to the midplane displacement and those due to the displacements normal to the midplane.
- These are called the **membrane strains** and the **bending (or flexural) strains**, respectively.

Next: Nonlinear Finite Element

Thanks for your attention!